



GITHUB ACTIONS: FROM ZERO TO HERO

A comprehensive, stage-by-stage guide – from your first workflow file to advanced CI/CD pipelines – built on 3 years of real-world experience. Whether you're automating your first test run or orchestrating multi-environment deployments, this guide will take you there.

BEGINNER → ADVANCED

CI/CD

AUTOMATION

WHAT WE'LL COVER

YOUR LEARNING JOURNEY

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What GitHub Actions is, why it matters, and the core vocabulary every developer needs to know

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Secrets, environments, reusable workflows, matrix builds, and real-world CI pipelines

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YOUR FIRST WORKFLOW

YAML syntax, triggers, jobs, steps – writing and running your first automation from scratch

04

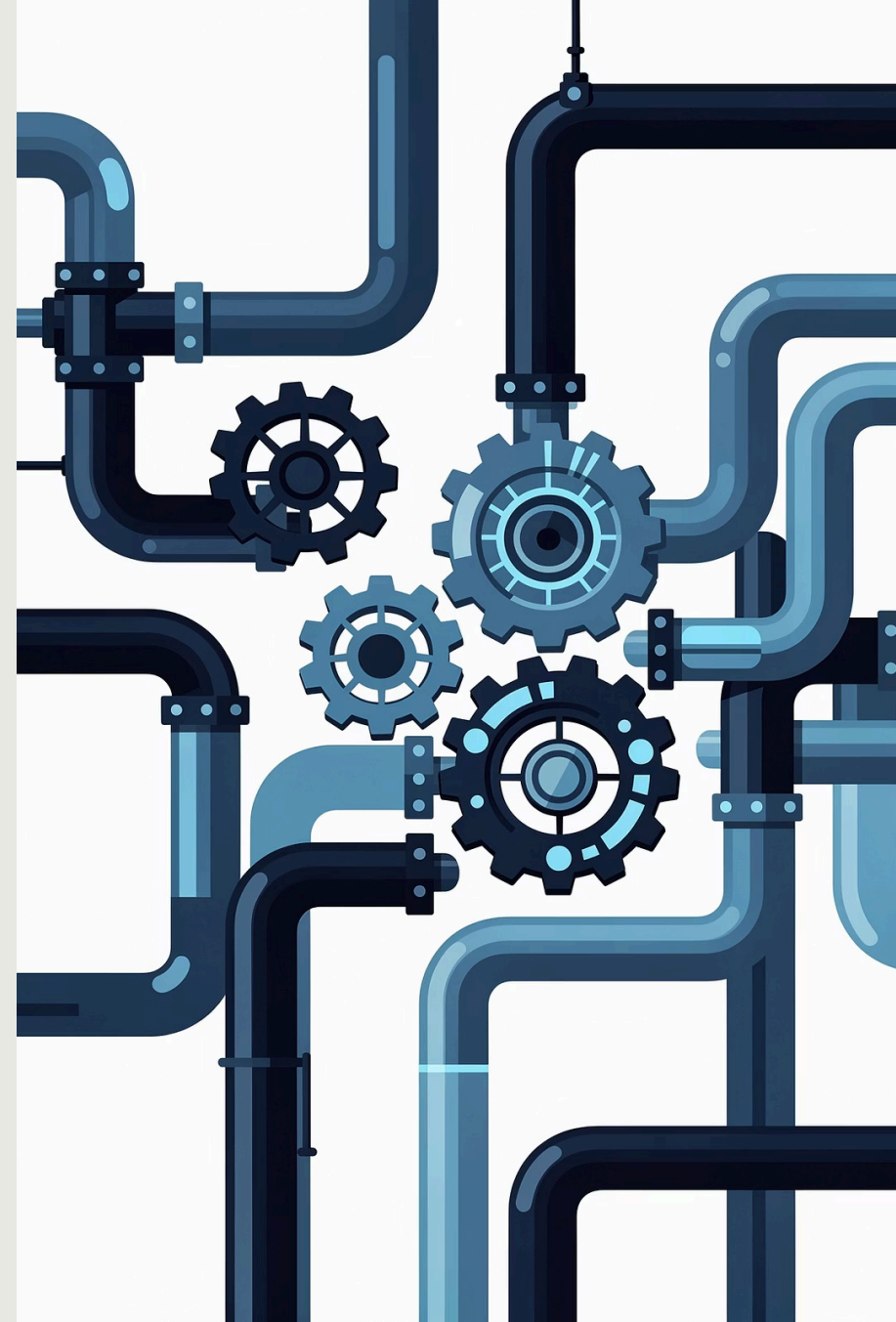
ADVANCED MASTERY

Custom actions, self-hosted runners, OIDC, concurrency control, and production-grade CD

WHAT IS GITHUB ACTIONS?

GitHub Actions is GitHub's built-in **CI/CD and automation platform**. It lets you automate software workflows – testing, building, deploying, and more – directly inside your GitHub repository, triggered by events like pushes, pull requests, or a schedule.

Introduced in 2018 and made generally available in November 2019, GitHub Actions has become one of the most widely-used automation platforms in the software industry, replacing tools like Jenkins, CircleCI, and TravisCI for millions of teams.



WHY GITHUB ACTIONS?

NATIVE INTEGRATION

Lives inside your repo — no third-party setup, no webhook plumbing required

MARKETPLACE

15,000+ community-built and official actions available to reuse instantly

EVENT-DRIVEN

Triggered by 35+ GitHub events — push, PR, issue, release, schedule, and more

GENEROUS FREE TIER

2,000 free minutes/month for private repos; unlimited for public repos

CORE VOCABULARY: THE BUILDING BLOCKS

Before writing a single line of YAML, you need to internalize these six terms. Every GitHub Actions concept maps back to them.

1

WORKFLOW

A YAML file in `.github/workflows/` that defines the entire automation. One repo can have many workflows.

2

EVENT

The trigger that starts a workflow – e.g., `push`, `pull_request`, `schedule`, `workflow_dispatch`

3

JOB

A group of steps that run on the same runner (virtual machine). Jobs run in parallel by default.

4

STEP

An individual task inside a job – either a shell command or a pre-built Action.

5

RUNNER

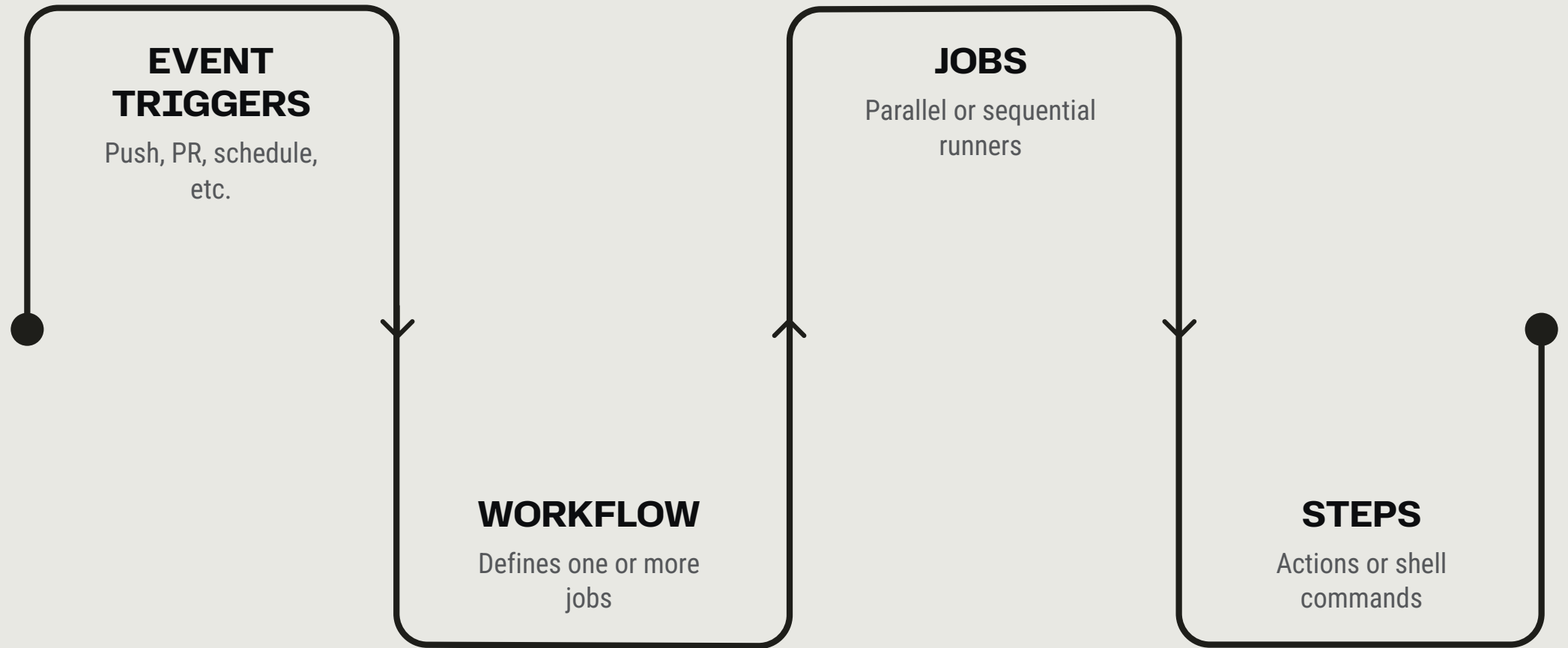
The virtual machine (Ubuntu, Windows, macOS) that executes the job.

6

ACTION

A reusable unit of work – like `actions/checkout` or `actions/setup-node`. Think of them as functions.

ARCHITECTURE OVERVIEW



A single GitHub event — like a push to `main` — fans out into one or more workflows, each containing parallel or sequential jobs, each job containing ordered steps. Understanding this hierarchy is the key to designing efficient pipelines that run fast and fail informatively.

WORKFLOW FILE ANATOMY

Every workflow is a `.yaml` file placed in `.github/workflows/`. Here is a fully annotated example of the minimal structure:

```
# .github/workflows/hello.yml

name: Hello World Workflow      # Displayed in the GitHub UI

on:                             # TRIGGER: when to run
  push:
    branches: [main]

jobs:                           # JOBS: what to run
  say-hello:                    # Job ID (snake_case)
    runs-on: ubuntu-latest      # RUNNER: which OS

    steps:                      # STEPS: ordered tasks
      - name: Print greeting
        run: echo "Hello, GitHub Actions!"
```

Save this file, push to `main`, then navigate to the **Actions** tab of your repo. You will see the workflow run in real time.

EVENTS (TRIGGERS) DEEP DIVE

The `on:` key is where you declare what causes your workflow to run. You can listen to one event, multiple events, or scheduled times using cron syntax.

PUSH / PULL REQUEST

`on: push` or `on: pull_request` – filter by branch or file path using `branches:` and `paths:`

SCHEDULE (CRON)

`on: schedule: - cron: '0 8 * * 1'` – runs every Monday at 8AM UTC. Great for nightly builds or dependency audits.

MANUAL TRIGGER

`on: workflow_dispatch` – adds a "Run workflow" button in the GitHub UI. Accepts custom input parameters.

REPOSITORY EVENTS

`on: release`, `on: issues`, `on: pull_request_review` – automate responses to project activities.

TRIGGER SYNTAX: PRACTICAL EXAMPLES

```
on:
  push:
    branches: [main, develop]    # Only on these branches
    paths:
      - 'src/**'                 # Only when src/ changes

  pull_request:
    types: [opened, synchronize] # PR opened or updated

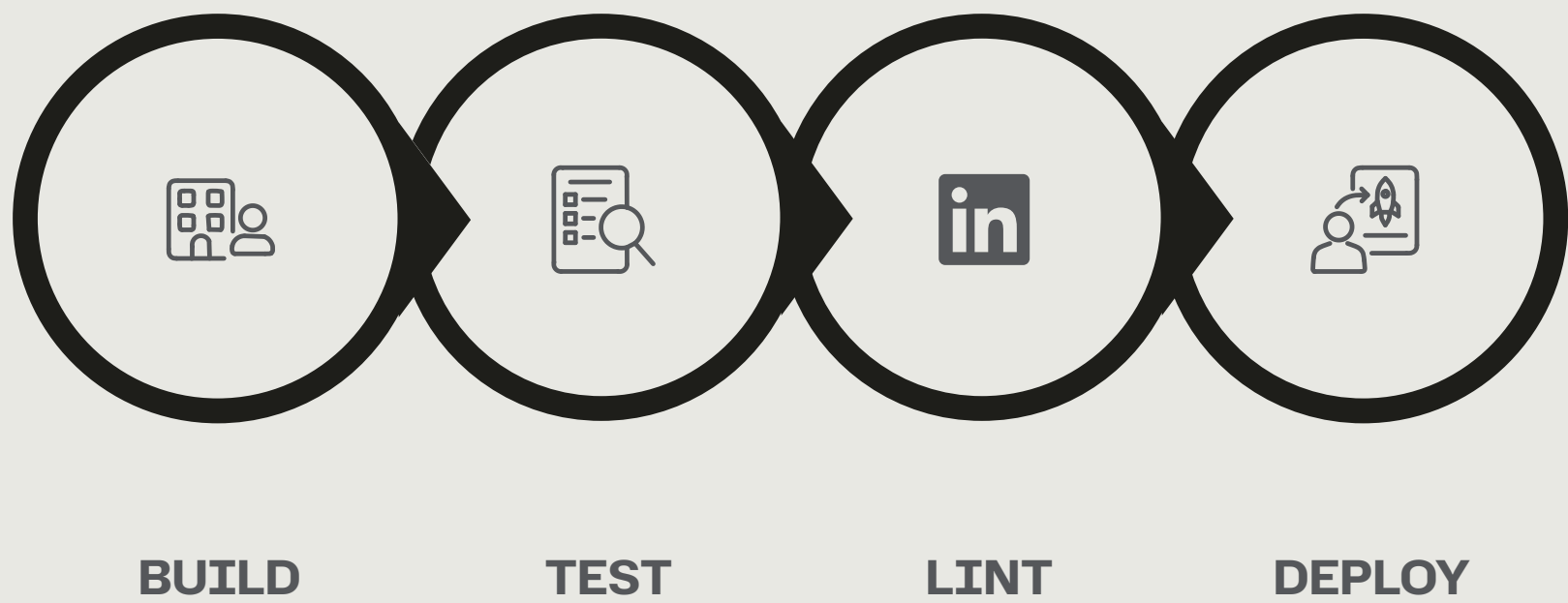
  schedule:
    - cron: '30 5 * * 1-5'       # Weekdays at 5:30 AM UTC

  workflow_dispatch:             # Manual run with input
    inputs:
      environment:
        description: 'Deploy target'
        required: true
        default: 'staging'
        type: choice
        options: [staging, production]
```

  **Pro tip:** Use paths-ignore to skip CI when only docs or README files change – this saves runner minutes.

JOBS: PARALLEL VS. SEQUENTIAL

By default, all jobs in a workflow run **in parallel**. Use the `needs:` keyword to create dependencies and force sequential execution. This is the foundation of a proper CI/CD pipeline.



```
jobs:
  build:
    runs-on: ubuntu-latest
    steps: [...]

  test:
    needs: build # Waits for build to succeed
    runs-on: ubuntu-latest
    steps: [...]

  lint:
    needs: build # Also waits for build
    runs-on: ubuntu-latest
    steps: [...]

  deploy:
    needs: [test, lint] # Waits for BOTH
    runs-on: ubuntu-latest
    steps: [...]
```



RUNNERS: YOUR EXECUTION ENVIRONMENT

A **runner** is the virtual machine that actually executes your job. GitHub provides hosted runners with pre-installed software, or you can self-host your own.

UBUNTU-LATEST

Most popular. Fast, cheap, great for Linux builds, Docker, and most Node/Python/Go workflows.

WINDOWS-LATEST

For .NET, PowerShell, or Windows-specific builds. Slower and uses more minutes.

MACOS-LATEST

Required for iOS/macOS app builds (Xcode). Uses 10× more minutes than Linux.

SELF-HOSTED

Your own machine or cloud VM. Unlimited minutes, custom hardware, but you manage maintenance.

STEPS: THE HEART OF EVERY JOB

Each step either runs a shell command (`run:`) or calls a pre-built Action (`uses:`). Steps execute in order and share the same filesystem within a job.

steps:

Step 1: Checkout your code (always do this first!)

- `name:` Checkout repository

`uses:` actions/checkout@v4

Step 2: Set up Node.js with a specific version

- `name:` Setup Node.js

`uses:` actions/setup-node@v4

`with:`

`node-version:` '20'

`cache:` 'npm' # Built-in caching!

Step 3: Install dependencies

- `name:` Install dependencies

`run:` npm ci

Step 4: Run tests

- `name:` Run tests

`run:` npm test

Step 5: Multi-line shell command

- `name:` Build and archive

`run:` |

npm run build

echo "Build complete!"

tar -czf dist.tar.gz dist/

ACTIONS: REUSABLE BUILDING BLOCKS

An Action is a self-contained, reusable automation component. You reference actions from three sources: the official GitHub Marketplace, your own repository, or a Docker image. Always pin to a specific version using `@v4` or a commit SHA for security.



MARKETPLACE ACTIONS

uses: `actions/checkout@v4` 15,000+
actions for everything from AWS deployment
to Slack notifications



LOCAL ACTIONS

uses: `./github/actions/my-action` Define
an action inside your own repo for team-
specific reuse



DOCKER ACTIONS

uses: `docker://alpine:3.18` Run any
Docker image as a step – great for custom
tooling

ENVIRONMENT VARIABLES & CONTEXTS

GitHub Actions provides rich context objects you can access inside your YAML. They contain metadata about the run, repository, actor, and more.

```
jobs:
  display-context:
    runs-on: ubuntu-latest
    steps:
      - name: Show context values
        run: |
          echo "Repo:   ${{ github.repository }}"
          echo "Branch: ${{ github.ref_name }}"
          echo "Actor:  ${{ github.actor }}"
          echo "Run ID: ${{ github.run_id }}"
          echo "SHA:   ${{ github.sha }}"

      - name: Set custom env var
        env:
          APP_ENV: production
          BUILD_VERSION: ${{ github.run_number }}
        run: |
          echo "Deploying version $BUILD_VERSION to $APP_ENV"
```

📄 📌 Key contexts: github (event metadata), env (variables), secrets (encrypted values), steps (previous step outputs), runner (OS info)

SECRETS: KEEPING CREDENTIALS SAFE

Never hardcode passwords, API keys, or tokens in your YAML. GitHub Secrets are encrypted at rest and masked in logs. Add them in **Settings** → **Secrets and variables** → **Actions**, then reference them with `${{ secrets.MY_SECRET }}`.

```
jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - name: Configure AWS credentials
        uses: aws-actions/configure-aws-credentials@v4
        with:
          aws-access-key-id:  ${{{ secrets.AWS_ACCESS_KEY_ID }}}
          aws-secret-access-key: ${{ secrets.AWS_SECRET_ACCESS_KEY }}
          aws-region:        us-east-1

      - name: Deploy to S3
        run: aws s3 sync ./dist s3://${{ secrets.S3_BUCKET }}
```

REPOSITORY SECRETS

Available to all workflows in a single repo. Best for project-specific keys.

ORGANIZATION SECRETS

Shared across multiple repos in an org. Ideal for shared service accounts.

ENVIRONMENT SECRETS

Scoped to a specific deployment environment (staging, production) with approval gates.

A COMPLETE CI PIPELINE EXAMPLE

Here is a full, production-ready CI workflow for a Node.js application — covering install, lint, test, and build in a single file:

```
name: CI Pipeline

on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

jobs:
  ci:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v4

      - uses: actions/setup-node@v4
        with:
          node-version: '20'
          cache: 'npm'

      - name: Install
        run: npm ci

      - name: Lint
        run: npm run lint

      - name: Test with coverage
        run: npm run test -- --coverage

      - name: Upload coverage report
        uses: actions/upload-artifact@v4
        with:
          name: coverage-report
          path: coverage/
          retention-days: 7

      - name: Build
        run: npm run build
```

MATRIX BUILDS: TEST ACROSS ENVIRONMENTS

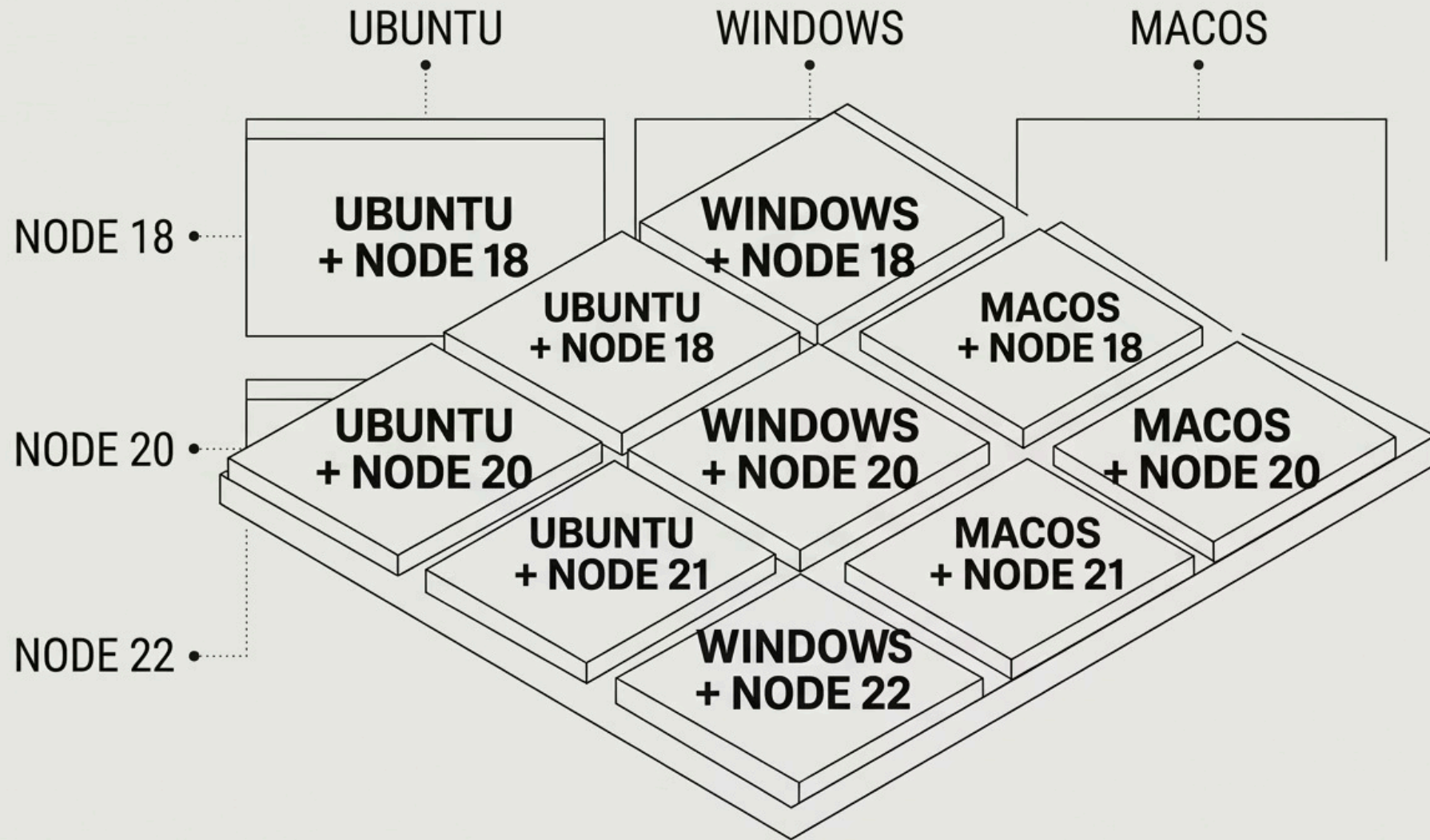
The `matrix` strategy lets you run a job multiple times with different variable combinations — ideal for testing across Node versions, operating systems, or database backends without duplicating code.

```
jobs:
  test-matrix:
    runs-on: ${{ matrix.os }}
    strategy:
      matrix:
        os: [ubuntu-latest, windows-latest, macos-latest]
        node-version: [18, 20, 22]
    fail-fast: false # Don't cancel all jobs if one fails
```

```
steps:
  - uses: actions/checkout@v4
  - uses: actions/setup-node@v4
    with:
      node-version: ${{ matrix.node-version }}
  - run: npm ci
  - run: npm test
```

This single job definition expands into **9 parallel runs** (3 OS × 3 Node versions), dramatically speeding up your compatibility testing and catching environment-specific bugs before they reach production.

MATRIX STRATEGY VISUALIZED



Each cell in the matrix becomes an independent job. They all run simultaneously, and the GitHub UI clearly shows which combination passed and which failed — making it easy to pinpoint environment-specific issues at a glance.

CACHING DEPENDENCIES

Without caching, every run re-downloads your entire `node_modules`, `pip`, or `maven` cache – wasting minutes and money. Use the built-in `cache` option in setup actions, or use `actions/cache` directly.

steps:

Method 1: Built-in cache (preferred for common runtimes)

- `uses: actions/setup-node@v4`

with:

`node-version: '20'`

`cache: 'npm'` # Caches ~/.npm automatically

Method 2: Manual cache for custom paths

- `uses: actions/cache@v4`

with:

`path: |`

`~/.npm`

`./.next/cache`

`key: ${{ runner.os }}-node-${{ hashFiles('**/package-lock.json') }}`

`restore-keys: |`

`${{ runner.os }}-node-`

  **Real-world impact:** Caching `node_modules` can reduce a 3-minute install step to under 10 seconds – a 95% improvement on cache hits.

ARTIFACTS: SHARING DATA BETWEEN JOBS

Jobs run on isolated runners and cannot share files directly. Use **artifacts** to upload outputs from one job and download them in another. Artifacts are also useful for saving test reports, build binaries, or logs for later inspection.

```
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - run: npm ci && npm run build
      - name: Upload build artifact
        uses: actions/upload-artifact@v4
    with:
      name: production-build
      path: dist/
      retention-days: 30

  deploy:
    needs: build
    runs-on: ubuntu-latest
    steps:
      - name: Download build artifact
        uses: actions/download-artifact@v4
    with:
      name: production-build
      path: dist/
      - name: Deploy
        run: ./scripts/deploy.sh dist/
```

REUSABLE WORKFLOWS

Reusable workflows eliminate copy-paste across repositories. Define a workflow once and call it from any other workflow – like a function call between YAML files. This is a game-changer for platform teams managing CI standards across dozens of repos.

DEFINING A REUSABLE WORKFLOW

```
# .github/workflows/reusable-deploy.yml
on:
  workflow_call:      # Makes it reusable
  inputs:
    environment:
      required: true
      type: string
  secrets:
    DEPLOY_KEY:
      required: true

jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - run: |
          echo "Deploying to ${ inputs.environment }"
```

CALLING THE REUSABLE WORKFLOW

```
# In ANY repo's workflow:
jobs:
  call-deploy:
    uses: my-org/shared-workflows/
      .github/workflows/
      reusable-deploy.yml@main
    with:
      environment: production
    secrets:
      DEPLOY_KEY: ${ secrets.DEPLOY_KEY }
```

The calling workflow delegates the entire job to the shared definition – keeping your CI/CD logic centralized and versioned.

ENVIRONMENTS & DEPLOYMENT GATES

GitHub Environments add protection rules to your deployments – requiring manual approval, limiting which branches can deploy, and scoping secrets to specific environments like staging or production.



STAGING

PRE-PRODUCTION

PRODUCTION

```
jobs:
  deploy-production:
    runs-on: ubuntu-latest
    environment:
      name: production # Links to a GitHub Environment
      url: https://myapp.com # Shown in the GitHub UI
    steps:
      - name: Deploy
        run: ./deploy.sh
    env:
      API_KEY: ${ secrets.PROD_API_KEY } # Env-scoped secret
```

CONDITIONAL STEPS & JOB CONTROL

Use the `if:` keyword to control whether a step or job runs based on context, outcome of previous steps, or custom conditions. This is essential for building smart pipelines that behave differently on different branches or outcomes.

```
jobs:
  smart-job:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Run tests
        id: tests
        run: npm test

      # Only runs if tests FAILED
      - name: Notify Slack on failure
        if: failure()
        uses: slackapi/slack-github-action@v1
        with:
          payload: '{"text":'
```

```
❌ Tests failed on ${github.ref_name}"'
        channel-id: 'engineering-alerts'
        env:
          SLACK_BOT_TOKEN: ${secrets.SLACK_BOT_TOKEN}
```

```
      # Only runs on main branch, even if prior steps failed
      - name: Create release tag
        if: github.ref == 'refs/heads/main' && success()
        run: gh release create v${github.run_number}
```

CONCURRENCY CONTROL

By default, multiple workflow runs can execute simultaneously. This can cause race conditions on deployments. The `concurrency` key lets you queue or cancel redundant runs — a critical setting for any deployment workflow.

```
# Cancel any in-progress run of the same workflow on the same branch
```

```
concurrency:
```

```
  group: deploy-${{ github.ref }}
```

```
  cancel-in-progress: true
```

```
jobs:
```

```
  deploy:
```

```
    runs-on: ubuntu-latest
```

```
    steps:
```

```
      - run: ./deploy.sh
```

CANCEL-IN-PROGRESS: TRUE

Cancels the running workflow when a newer one starts. Best for feature branches where only the latest matters.

CANCEL-IN-PROGRESS: FALSE

Queues new runs, waiting for the current one to finish. Best for production deploys where order matters.

BUILDING A CUSTOM ACTION

When no marketplace action fits your needs, build your own. Custom actions can be written in JavaScript/TypeScript (node20) or as a Composite action (pure YAML shell steps). Here is a minimal JavaScript action:

```
# .github/actions/greet/action.yml
name: 'Greet User'
description: 'Prints a personalized greeting'
inputs:
  username:
    description: 'GitHub username to greet'
    required: true
outputs:
  greeting:
    description: 'The greeting message'

runs:
  using: 'node20'
  main: 'index.js'
```

```
// .github/actions/greet/index.js
const core = require('@actions/core');
try {
  const username = core.getInput('username');
  const greeting = `Hello, ${username}! Welcome to CI.`;
  core.setOutput('greeting', greeting);
  console.log(greeting);
} catch (error) {
  core.setFailed(error.message);
}
```

COMPOSITE ACTIONS: YAML-ONLY ACTIONS

Composite actions bundle multiple steps into a single reusable unit without writing JavaScript. They're the easiest way to encapsulate repetitive steps – like a standard "setup and install" block – and share them across multiple workflows.

```
# .github/actions/setup-and-install/action.yml
name: 'Setup Node and Install'
description: 'Checkout, setup Node, and install dependencies'
inputs:
  node-version:
    description: 'Node.js version'
    default: '20'

runs:
  using: 'composite'
  steps:
    - uses: actions/checkout@v4

    - uses: actions/setup-node@v4
      with:
        node-version: '${{ inputs.node-version }}'
        cache: 'npm'

    - name: Install dependencies
      run: npm ci
      shell: bash
```

  Use composite actions to enforce a consistent "setup ritual" across all repos in your organization – one update to the action propagates everywhere.



SELF-HOSTED RUNNERS

GitHub-hosted runners are convenient but have limitations: 6-hour job limit, no custom hardware, and costs scale with usage. Self-hosted runners run on *your* infrastructure – a VM, Kubernetes pod, or bare-metal server – and give you full control.

WHEN TO USE SELF-HOSTED

Large build artifacts, GPU workloads, private network access, compliance requirements, or need for custom software pre-installed

SETUP IN 3 COMMANDS

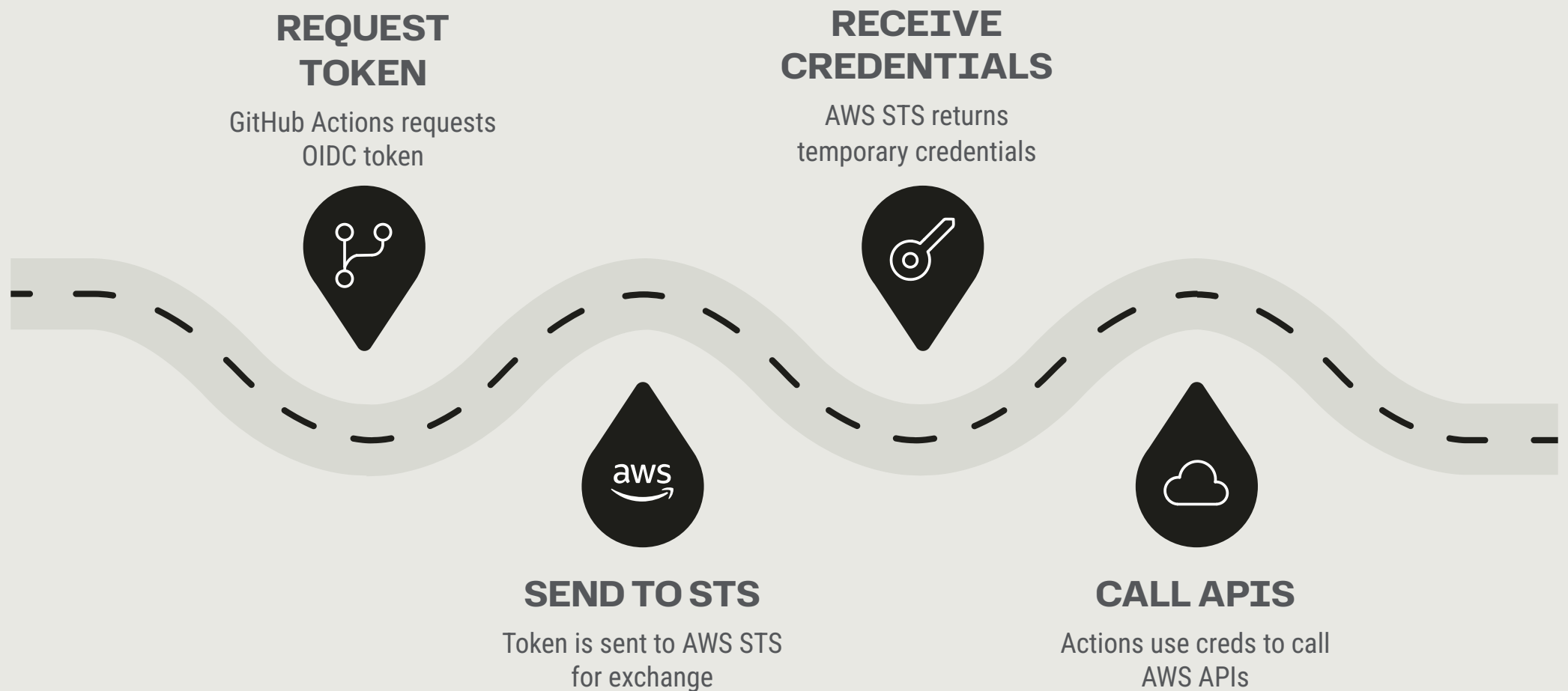
Download the runner agent, configure with your repo token, then `./run.sh`. Reference with `runs-on: self-hosted`

RUNNER LABELS

Tag runners with labels like `gpu`, `arm64`, or `high-memory` and target them with `runs-on: [self-hosted, gpu]`

OIDC: KEYLESS CLOUD AUTHENTICATION

Storing long-lived cloud credentials (AWS keys, GCP service account JSON) as secrets is a security risk. OpenID Connect (OIDC) lets GitHub Actions request short-lived tokens from your cloud provider on-the-fly — no stored credentials needed.



```
# Grant OIDC permission at the top of your workflow
```

```
permissions:
```

```
id-token: write
```

```
contents: read
```

```
jobs:
```

```
deploy:
```

```
runs-on: ubuntu-latest
```

```
steps:
```

```
- name: Authenticate to AWS via OIDC
```

```
uses: aws-actions/configure-aws-credentials@v4
```

```
with:
```

```
role-to-assume: arn:aws:iam::123456789:role/GitHubActionsRole
```

```
aws-region: us-east-1
```

```
# No secrets needed! Token is generated automatically
```

SECURITY HARDENING BEST PRACTICES

GitHub Actions is powerful, but that power can be abused. Following these practices is essential before shipping to production environments.

→ PIN ACTIONS TO A COMMIT SHA

Use `uses:`

`actions/checkout@11bd71901bbe5b1630ceea73d27597364c9af`

`683` instead of `@v4` to prevent supply-chain attacks from a compromised tag.

→ MINIMIZE PERMISSIONS

Add `permissions: read-all` at the workflow level, then grant only what each job needs. Avoid `write-all`.

→ NEVER ECHO SECRETS

Secrets are masked in logs, but `run: echo ${ secrets.MY_KEY }` can expose them in forked PRs. Use environment variables instead.

→ RESTRICT `WORKFLOW_DISPATCH`

Protect manual triggers with environment rules and branch protection. Don't allow arbitrary branches to deploy to production.

DEBUGGING FAILED WORKFLOWS

When a workflow fails, GitHub's log viewer is your first stop. But for deeper debugging, enable step debug logging or SSH into a running runner using a special action.

```
# Enable step debug logging:
# Add this as a repository secret:
# ACTIONS_STEP_DEBUG = true
# ACTIONS_RUNNER_DEBUG = true

# Then add tmate for interactive SSH debugging:
steps:
  - uses: actions/checkout@v4
  - name: Run tests
    run: npm test

# Add this AFTER the failing step to pause and SSH in
- name: Setup tmate session (debug only)
  if: failure()      # Only runs if previous step failed
  uses: mxschmitt/action-tmate@v3
  timeout-minutes: 15 # Auto-closes after 15 min
```

  **Workflow run logs** are available for 90 days by default. Navigate to Actions → select the run → click any job → expand any step to see its full output.

COMPLETE CD PIPELINE: DEPLOY TO AWS ECS

Here is a real-world, end-to-end CD pipeline that builds a Docker image, pushes it to Amazon ECR, and deploys to ECS – the kind of workflow you'd use in production.

```
name: Deploy to Production

on:
  push:
    branches: [main]

permissions:
  id-token: write
  contents: read

jobs:
  deploy:
    runs-on: ubuntu-latest
    environment: production

    steps:
      - uses: actions/checkout@v4

      - name: Authenticate to AWS
        uses: aws-actions/configure-aws-credentials@v4
        with:
          role-to-assume: ${{ secrets.AWS_ROLE_ARN }}
          aws-region: us-east-1

      - name: Login to Amazon ECR
        id: ecr-login
        uses: aws-actions/amazon-ecr-login@v2

      - name: Build & push Docker image
        env:
          REGISTRY: ${{ steps.ecr-login.outputs.registry }}
          IMAGE_TAG: ${{ github.sha }}
        run: |
          docker build -t $REGISTRY/myapp:$IMAGE_TAG .
          docker push $REGISTRY/myapp:$IMAGE_TAG

      - name: Deploy to ECS
        uses: aws-actions/amazon-ecs-deploy-task-definition@v2
        with:
          task-definition: task-def.json
          service: myapp-service
          cluster: production-cluster
          wait-for-service-stability: true
```

WORKFLOW DISPATCH WITH PARAMETERS

Manual workflows are extremely useful for one-off tasks: database migrations, feature flag toggles, manual rollbacks. The `workflow_dispatch` event supports rich input types that render as a UI form in GitHub.

```
name: Manual Deploy

on:
  workflow_dispatch:
inputs:
  environment:
    description: 'Target environment'
    required: true
    type: choice
    options: [staging, production]
  version:
    description: 'Image tag to deploy'
    required: true
    type: string
  dry_run:
    description: 'Dry run only (no actual deploy)'
    required: false
    type: boolean
    default: false

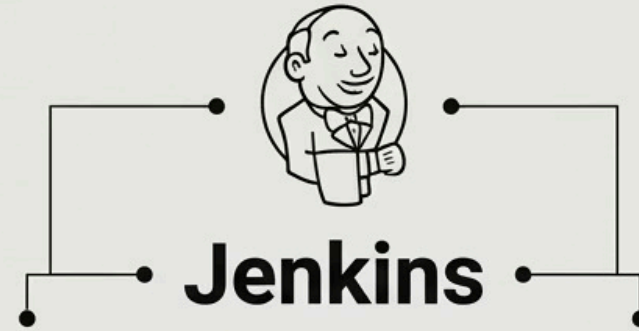
jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - name: Deploy
        run: |
          if [ "${{ inputs.dry_run }}" = "true" ]; then
            echo "DRY RUN: Would deploy ${{ inputs.version }} to ${{ inputs.environment }}"
          else
            ./deploy.sh ${{ inputs.environment }} ${{ inputs.version }}
          fi
```

GITHUB ACTIONS VS. OTHER CI/CD TOOLS



GitHub Actions

Native integration • YAML config public repos



Jenkins

Self-hosted • Plugin ecosystem • Full control



CircleCI

Fast, parallel testing • Orbs (pre-built config)
Orbs (pre-built config) • Docker-first



GitLab CI

Integrated with GitLab • Scalable runners
Integration • Part of DevOps platform

GitHub Actions wins on integration and convenience for GitHub-native teams. Jenkins offers the most flexibility for complex enterprise environments. The right choice always depends on your team's existing infrastructure and workflow complexity.

REAL-WORLD PIPELINE ARCHITECTURE



PUSH CODE

CI: LINT & TEST

BUILD & SCAN

STAGING DEPLOY

This is the gold-standard pipeline pattern used by high-performing engineering teams. Every stage adds a layer of confidence: unit tests catch logic bugs, security scans catch vulnerabilities, staging validates integration, and approval gates ensure human judgment before production changes go live.

TIPS FROM 3 YEARS OF PRODUCTION USE



KEEP JOBS FAST

Target under 5 minutes per job. Split long-running jobs into parallel ones. Use caching aggressively. Slow CI is abandoned CI.



TREAT WORKFLOW FILES AS CODE

Review workflow changes in PRs. Require approval for changes to deploy workflows. Version your reusable workflows with tags, not `@main`.



INVEST IN REUSABILITY EARLY

Build composite or reusable workflows from the start. The 3rd time you copy-paste a step is the last time — abstract it.



MONITOR YOUR MINUTE USAGE

Use `paths-ignore`, skip jobs on docs-only changes, and avoid macOS runners unless absolutely necessary. Costs add up fast.

OUTPUTS: PASSING DATA BETWEEN STEPS

Steps can produce outputs that subsequent steps consume using the `steps` context. This is essential for passing dynamic values – like a generated version number, a computed tag, or the result of a script – between steps within the same job.

```
jobs:
  versioning:
    runs-on: ubuntu-latest
    steps:
      - name: Generate version
        id: version      # Give this step an ID
        run: |
          VERSION="1.0.${{ github.run_number }}"
          echo "tag=$VERSION" >> $GITHUB_OUTPUT # Write output

      - name: Use the version
        run: |
          echo "Building version ${{ steps.version.outputs.tag }}"
          docker build -t myapp:${{ steps.version.outputs.tag }} .

      - name: Create Git tag
        run: |
          git tag ${{ steps.version.outputs.tag }}
          git push origin ${{ steps.version.outputs.tag }}
```

📄 📌 Always use `$GITHUB_OUTPUT` to set outputs. The old `set-output` command was deprecated in 2022 and will fail in modern runners.

JOB OUTPUTS: PASSING DATA BETWEEN JOBS

While step outputs stay within a job, **job outputs** let you share values across jobs in the same workflow. The pattern requires declaring outputs at the job level and consuming them via the `needs` context.

```
jobs:
  build:
    runs-on: ubuntu-latest
    outputs:
      image-tag: ${ steps.tag.outputs.value } # Declare job output

    steps:
      - name: Compute tag
        id: tag
        run: echo "value=sha-${ github.sha }" >> $GITHUB_OUTPUT

  deploy:
    needs: build
    runs-on: ubuntu-latest
    steps:
      - name: Deploy image
        run: |
          # Consume the output from the 'build' job
          echo "Deploying image tag: ${ needs.build.outputs.image-tag }"
          ./deploy.sh ${ needs.build.outputs.image-tag }
```

This pattern is the backbone of any pipeline that computes a version or artifact name in one job and needs it in multiple downstream jobs — like deploying the same Docker tag to staging and then production.

EXPRESSIONS & FUNCTIONS

GitHub Actions expressions (`${{ }}`) support a rich set of built-in functions for string manipulation, JSON parsing, comparisons, and status checks. Mastering these unlocks highly dynamic workflows.

STATUS FUNCTIONS

`success()`, `failure()`, `cancelled()`, `always()` – used in `if:` to respond to job outcomes conditionally

STRING FUNCTIONS

`contains()`, `startsWith()`, `endsWith()`, `format()`, `join()` – manipulate strings inside expressions

JSON FUNCTIONS

`fromJson()` and `toJson()` – parse JSON strings into objects or serialize objects. Critical for dynamic matrix generation.

HASH FUNCTIONS

`hashFiles('**/package-lock.json')` – generates a hash of file contents, perfect for building deterministic cache keys.

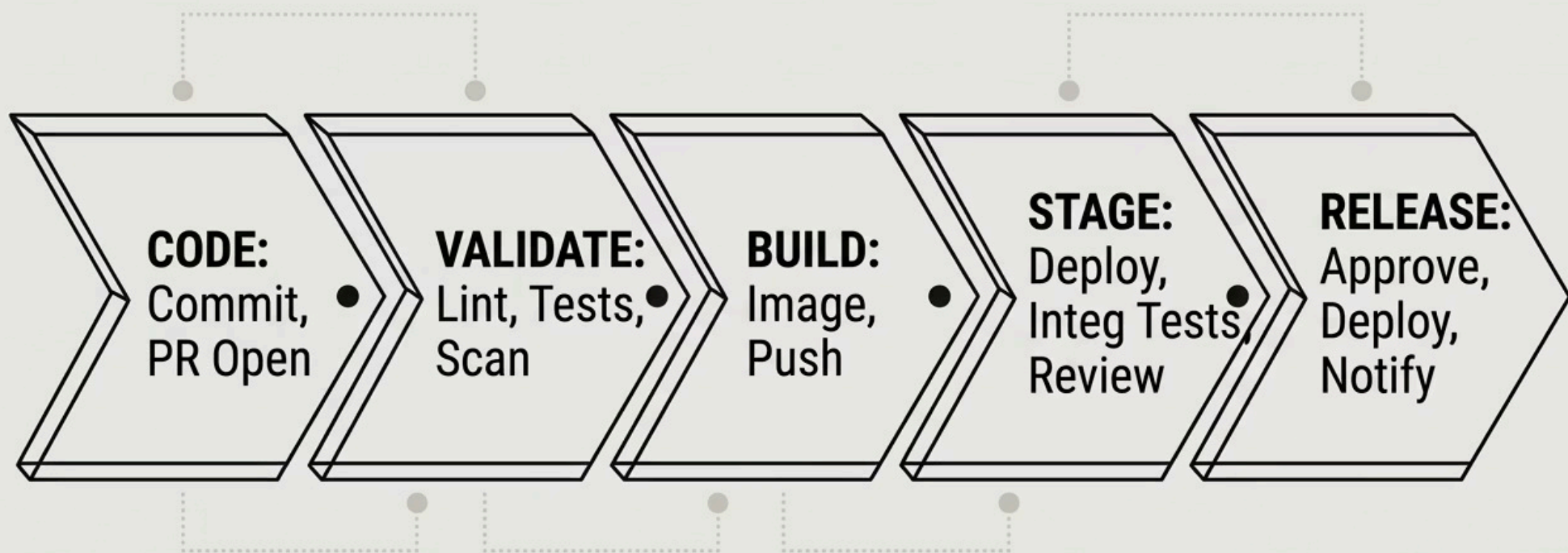
DYNAMIC MATRIX GENERATION

Advanced teams generate matrix values *dynamically* at runtime rather than hardcoding them. This enables matrices that reflect the actual state of your repo – for example, discovering which microservices changed and only testing those.

```
jobs:
  # Step 1: Compute the matrix dynamically
  setup-matrix:
    runs-on: ubuntu-latest
    outputs:
      matrix: ${ steps.set-matrix.outputs.matrix }
    steps:
      - uses: actions/checkout@v4
      - name: Detect changed services
      id: set-matrix
      run: |
        # Find directories that changed vs main
        SERVICES=$(git diff --name-only origin/main |
        grep '^services/' |
        cut -d/ -f2 | sort -u |
        jq -R -s -c 'split("\n")[:-1]')
        echo "matrix={\"service\":$SERVICES}" >> $GITHUB_OUTPUT

  # Step 2: Use the dynamic matrix
  test:
    needs: setup-matrix
    strategy:
      matrix: ${ fromJSON(needs.setup-matrix.outputs.matrix) }
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - run: cd services/${ matrix.service } && npm test
```

WORKFLOW VISUALIZATION: FULL CI/CD LIFECYCLE



This lifecycle represents the mature GitHub Actions pattern used by platform engineering teams at scale. Each phase is independently triggerable, observable, and rollback-able – giving teams both speed and safety in their delivery process.

USING GITHUB_ENV & GITHUB_PATH

Beyond step outputs, GitHub provides two special file-based environment mechanisms that persist across steps within a job: `GITHUB_ENV` for setting environment variables, and `GITHUB_PATH` for adding to the system PATH.

steps:

- name: Set environment variables **for** all subsequent steps

run: |

```
echo "APP_VERSION=2.5.${{ github.run_number }}" >> $GITHUB_ENV
```

```
echo "DEPLOY_REGION=us-east-1" >> $GITHUB_ENV
```

```
echo "BUILD_TIMESTAMP=$(date -u +%Y%m%d%H%M%S)" >> $GITHUB_ENV
```

- name: Use variables **set** above

run: |

```
echo "Deploying version $APP_VERSION"
```

```
echo "Region: $DEPLOY_REGION"
```

```
echo "Built at: $BUILD_TIMESTAMP"
```

- name: Add custom tool to PATH

run: |

```
mkdir -p /opt/my-tools
```

```
cp ./scripts/my-cli /opt/my-tools/
```

```
echo "/opt/my-tools" >> $GITHUB_PATH
```

- name: Use the custom tool (now on PATH)

run: my-cli --version

CONTAINER JOBS & SERVICE CONTAINERS

GitHub Actions can spin up Docker containers alongside your job for integration testing – a PostgreSQL database, Redis cache, or any service your tests depend on. These are called **service containers** and they run for the lifetime of the job.

```
jobs:
  integration-tests:
    runs-on: ubuntu-latest

    services:
      postgres:
        image: postgres:15
        env:
          POSTGRES_USER: test
          POSTGRES_PASSWORD: testpass
          POSTGRES_DB: testdb
        ports:
          - 5432:5432
        options: >-
          --health-cmd pg_isready
          --health-interval 10s
          --health-timeout 5s
          --health-retries 5

      redis:
        image: redis:7-alpine
        ports:
          - 6379:6379

    steps:
      - uses: actions/checkout@v4
      - name: Run integration tests
        env:
          DATABASE_URL: postgres://test:testpass@localhost:5432/testdb
          REDIS_URL: redis://localhost:6379
        run: npm run test:integration
```

RUNNING JOBS INSIDE CONTAINERS

Instead of installing tools on the runner, you can run your entire job *inside* a Docker container. This gives you a perfectly reproducible environment and eliminates "works on my machine" issues across teams and runner versions.

```
jobs:
  build-with-container:
    runs-on: ubuntu-latest
    container:
      image: node:20-alpine      # Job runs INSIDE this container
      env:
        NODE_ENV: production
      options: --cpus 2          # Docker run options

  steps:
    - uses: actions/checkout@v4
    - name: Install dependencies
      run: npm ci                # Runs inside node:20-alpine
    - name: Build
      run: npm run build
    - name: Test
      run: npm test
```

  **Best practice:** Pin your container image to a specific digest (e.g., node:20-alpine@sha256:abc123...) in production workflows for full reproducibility and supply chain security.

DEPENDENCY REVIEW & SECURITY SCANNING

GitHub Actions integrates natively with GitHub's security features – dependency review, code scanning via CodeQL, and secret scanning. These workflows run automatically on PRs and surface vulnerabilities before they merge.

```
name: Security Scanning
```

```
on: [push, pull_request]
```

```
jobs:
```

```
  codeql:
```

```
    runs-on: ubuntu-latest
```

```
    permissions:
```

```
      security-events: write
```

```
      contents: read
```

```
  steps:
```

```
    - uses: actions/checkout@v4
```

```
    - name: Initialize CodeQL
```

```
      uses: github/codeql-action/init@v3
```

```
      with:
```

```
        languages: javascript, python
```

```
    - name: Autobuild
```

```
      uses: github/codeql-action/autobuild@v3
```

```
    - name: Perform CodeQL Analysis
```

```
      uses: github/codeql-action/analyze@v3
```

MONITORING WORKFLOW HEALTH

In production, you need visibility into how your workflows are performing over time – run duration trends, failure rates, and flaky tests. Use GitHub's native insights or export metrics to your observability stack.



GITHUB INSIGHTS

Navigate to **Actions** → **Usage** to see minute consumption by workflow. Identify expensive workflows and optimize them first.



FAILURE NOTIFICATIONS

Use Slack, Teams, or email actions in an `on: workflow_run` workflow to alert on failures across your org's most critical pipelines.



WORKFLOW RUN HISTORY

The Actions tab retains 90 days of run history. Filter by branch, status, or actor to investigate recurring failure patterns.



EXPORT TO OBSERVABILITY

Use the GitHub API or `workflow_run` events to push duration and status metrics to Datadog, Grafana, or CloudWatch for long-term trending.

COMMON MISTAKES TO AVOID

These are the most frequent pitfalls – collected from real teams – that cause security incidents, broken pipelines, or runaway cloud bills.

USING @MAIN FOR ACTIONS

A compromised action tag can inject malicious code. Always pin to a commit SHA or a released version tag like @v4.

SKIPPING NEEDS: ON DEPLOYS

A deploy job that doesn't need: [test] can deploy broken code. Always make your deploy depend on a passing CI job.

SECRETS IN RUN COMMANDS

Interpolating secrets directly into run: scripts (e.g., `curl -H "Auth: ${secrets.TOKEN}"`) can leak them in forked PR logs.

NO TIMEOUT ON JOBS

A hung job will consume minutes for up to 6 hours. Always set timeout-minutes: 15 (or appropriate) on every job.

WORKFLOW TIMEOUT & RETRY PATTERNS

Flaky tests and transient network failures are a reality. Use timeouts to prevent runaway jobs and retries to handle transient failures gracefully – without requiring manual re-runs.

```
jobs:
  resilient-job:
    runs-on: ubuntu-latest
    timeout-minutes: 20      # Kill job if it runs over 20 min

    steps:
      - uses: actions/checkout@v4

      # Retry a flaky step up to 3 times
      - name: Run flaky integration test
        uses: nick-fields/retry@v3
        with:
          timeout_minutes: 5
          max_attempts: 3
          retry_wait_seconds: 10
          command: npm run test:integration

      # Built-in continue-on-error for non-critical steps
      - name: Upload optional telemetry
        continue-on-error: true # Workflow succeeds even if this fails
        run: ./scripts/send-metrics.sh
```

📄 ⌚ Set timeout-minutes at both the job and step level for defense in depth. A job-level timeout catches everything; step-level timeouts give faster feedback on exactly which step hung.

GITHUB ACTIONS FOR RELEASE AUTOMATION

Automate your entire release process – changelog generation, GitHub Release creation, versioning, and package publishing – triggered by a tag push or manual dispatch. This eliminates error-prone manual release steps.

```
name: Release
```

```
on:
```

```
  push:
```

```
  tags:
```

```
    - 'v*.*.*' # Triggers on tags like v1.2.3
```

```
jobs:
```

```
  release:
```

```
    runs-on: ubuntu-latest
```

```
    permissions:
```

```
      contents: write # Needed to create releases
```

```
      packages: write # Needed to publish packages
```

```
  steps:
```

```
    - uses: actions/checkout@v4
```

```
    with:
```

```
      fetch-depth: 0 # Full history for changelog
```

```
    - name: Generate changelog
```

```
      id: changelog
```

```
      uses: conventional-changelog-action@v5
```

```
      with:
```

```
        github-token: ${ secrets.GITHUB_TOKEN }
```

```
    - name: Create GitHub Release
```

```
      uses: softprops/action-gh-release@v2
```

```
      with:
```

```
        body: ${ steps.changelog.outputs.clean_changelog }
```

```
        generate_release_notes: true
```

```
        files: |
```

```
          dist/*.tar.gz
```

```
          dist/*.zip
```

THE `GITHUB_TOKEN` SECRET

Every workflow run automatically receives a `GITHUB_TOKEN` – a temporary, scoped token that GitHub generates for each run. It can interact with the GitHub API (create issues, comment on PRs, push commits) without any manual configuration.

1

AUTO-GENERATED

Created at the start of each workflow run and expired when the run ends. Never stored or reused between runs.

2

SCOPED PERMISSIONS

Defaults to read on most resources. Grant additional write permissions using the `permissions:` key at the workflow or job level.

3

API ACCESS

Use it with `gh` CLI, `octokit`, or direct REST calls: `Authorization: Bearer ${secrets.GITHUB_TOKEN}`

4

LIMITS APPLY

Subject to GitHub API rate limits (5,000 requests/hour). For high-volume automation, use a dedicated GitHub App token instead.

ADVANCED: CALLING EXTERNAL APIS & WEBHOOKS

GitHub Actions workflows can reach out to any external API – triggering a Jira ticket, updating a deployment tracker, posting to Slack, or calling a custom internal service. Here's a clean pattern using `curl` and the `gh` CLI.

```
steps:
- name: Notify deployment tracker
  run: |
    curl -s -X POST \
    -H "Authorization: Bearer ${ secrets.DEPLOY_TRACKER_TOKEN }" \
    -H "Content-Type: application/json" \
    -d '{
      "service": "myapp",
      "version": "${ github.sha }",
      "environment": "production",
      "actor": "${ github.actor }",
      "timestamp": "$(date -u +%Y-%m-%dT%H:%M:%SZ)"
    }' \
    https://deploytracker.internal/api/deployments

- name: Comment on PR with deploy URL
  if: github.event_name == 'pull_request'
  env:
    GH_TOKEN: ${ secrets.GITHUB_TOKEN }
  run: |
    gh pr comment ${ github.event.pull_request.number } \
    --body "
```

✓ Deployed to staging: `https://pr-${ github.event.pull_request.number }.staging.myapp.com`

MASTERY CHECKLIST: ARE YOU PRODUCTION-READY?

Use this checklist to evaluate the maturity of your GitHub Actions setup. Teams that check all these boxes ship faster, safer, and with confidence.

CI RUNS ON EVERY PR, BLOCKING MERGE ON FAILURE

Branch protection rules enforce that tests, lint, and security scans pass before any code merges to `main`.

NO LONG-LIVED CREDENTIALS STORED AS SECRETS

OIDC is used for cloud providers; `GITHUB_TOKEN` handles GitHub API calls. Secrets rotate automatically or are short-lived.

REUSABLE WORKFLOWS SHARED ACROSS REPOS

Your platform team maintains a shared workflow repo. Application teams call standardized CI/CD workflows rather than rolling their own.

ALL JOBS HAVE TIMEOUT-MINUTES SET

No runaway jobs silently consuming minutes for hours. Every job has a sensible maximum duration configured.

DEPLOYMENT TO PRODUCTION REQUIRES HUMAN APPROVAL

GitHub Environments with required reviewers ensure no automated process can deploy to production without explicit sign-off.

QUICK REFERENCE: KEY SYNTAX PATTERNS

A fast-reference card for the most commonly used GitHub Actions patterns – bookmark this slide.

TRIGGERS

```
on:  
  push:  
    branches: [main]  
  pull_request:  
  schedule:  
    - cron: '0 9 * * 1'  
  workflow_dispatch:
```

CONDITIONALS

```
if: github.ref == 'refs/heads/main'  
if: failure()  
if: success() && !cancelled()  
if: contains(github.event.head_commit.message, '[skip ci])
```

PERMISSIONS

```
permissions:  
  contents: read  
  id-token: write  
  pull-requests: write  
  security-events: write
```

OUTPUTS & ENV

```
echo "key=value" >> $GITHUB_OUTPUT  
echo "VAR=value" >> $GITHUB_ENV  
echo "/path/to/bin" >> $GITHUB_PATH  
${{ steps.id.outputs.key }}  
${{ needs.job.outputs.key }}
```



LEARNING PATH & NEXT STEPS

You've covered the full spectrum – from your first `echo` step to OIDC, dynamic matrices, and production CD pipelines. Here is your recommended path forward to cement mastery.



BUILD YOUR FIRST REAL PIPELINE

Pick a personal project. Add CI (lint + test + build) triggered on PRs. Make it block merges.



WRITE A COMPOSITE ACTION

Extract your "setup and install" steps into a local composite action and reuse it across three workflows.



ADD A REAL DEPLOYMENT

Deploy to a cloud service using OIDC – no stored credentials. Add an Environment with a required reviewer.



EXPLORE THE MARKETPLACE

Browse github.com/marketplace?type=actions. Find five actions relevant to your stack and integrate them.

KEY TAKEAWAYS



MASTER THE HIERARCHY

Event → Workflow → Job → Step. Every GitHub Actions concept maps to this four-level hierarchy. Understand it deeply.



SECURITY IS NOT OPTIONAL

Pin actions, use OIDC, minimize permissions, and use environments with approval gates before you ship to production.



SPEED THROUGH REUSE

Caching, composite actions, reusable workflows, and matrix builds are the levers that keep CI fast and maintainable at scale.



AUTOMATE EVERYTHING

Testing, building, deploying, releasing, notifying – if a human does it more than twice, GitHub Actions can do it better.